

Assessing the suitability of reconstructed high-resolution hyperspectral imagery from PlanetScope and EMIT for plant functional trait estimation

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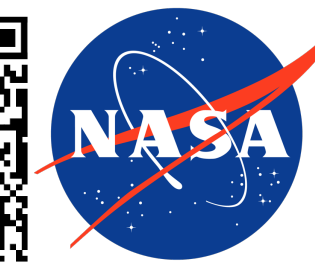
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About me



Code and Data



Motivation

Background:

- Plant traits at fine spatial scales are crucial for ecosystem and biodiversity applications.
- Satellite cannot capture both high spectral and spatial resolution observations simultaneously.
- Fusion-based HR-HSI offers a solution, but its value for trait estimation remains unknown.

Research questions:

- How much do reconstructed HR-HSI products improve plant trait estimation compared with EMIT (LR-HSI) and PlanetScope (HR-MSI)?
- Does embedding trait-specific spectral features into fusion model training enhance HR-HSI reconstruction and trait prediction?

Methods

Research area

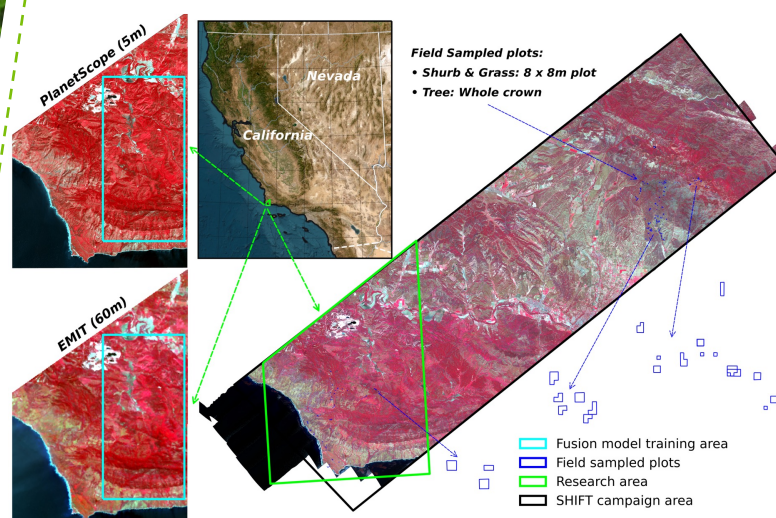


Fig. 1 Research area within NASA's SHIFT campaign region.

Remote Sensing data

- LR-HSI image: EMIT L2A**
381 – 2493 nm (7.5 nm) 60 m
Swath width: 75 km 285 bands
- HR-MSI image: PlanetScope L3B**
Coastal blue to NIR 3 m
Swath width: 32.5x19.6 km 8 bands
- HR-HSI image: AVIRIS-NG**
380 – 2510 nm (5 nm) 5 m
SHIFT campaign 425 bands

Modeling methods

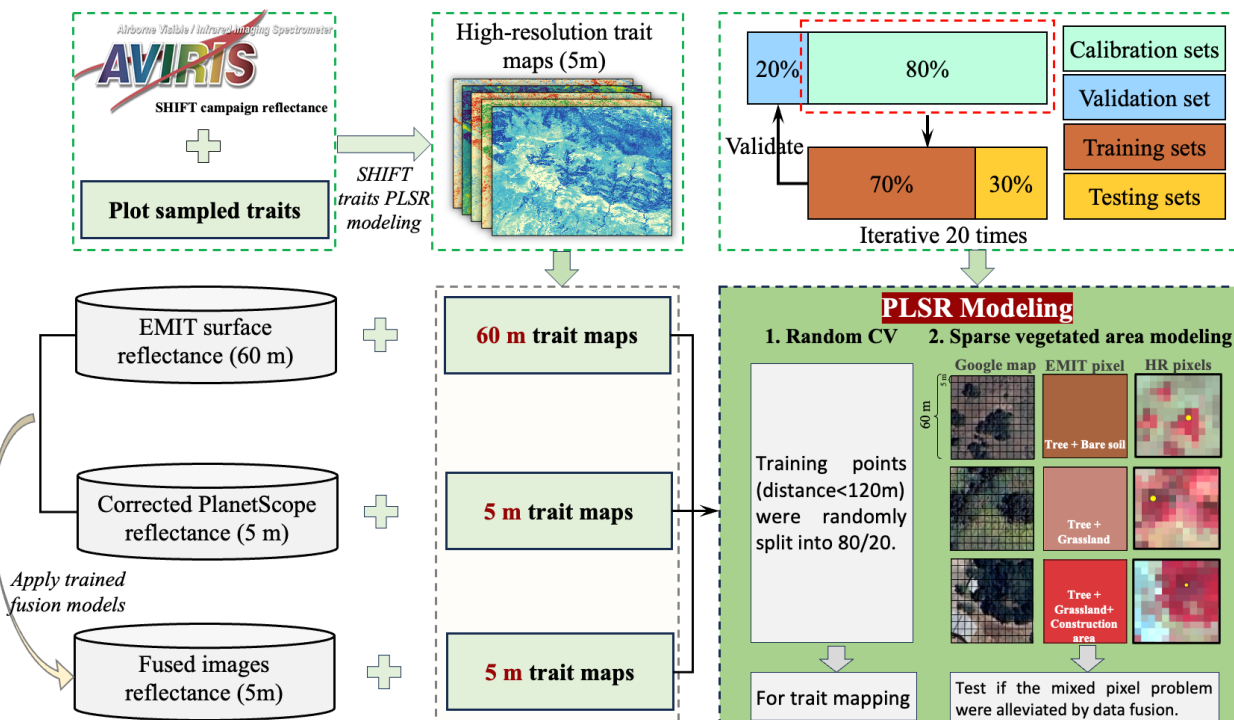


Fig. 2 Schematic representation of the workflow of this study.

We acknowledge the financial support from NASA Commercial Smallsat Data Scientific Analysis (CSDSA) Program.

Results

Trait mapping accuracy

Reflectance sources	EMIT	PlanetScope	DespNav	MSHNet	MSAHNet	ConSSFCNN	SSRNet
Chla+b (µg·cm ⁻²)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
LWC (mol·m ⁻²)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
LDMC (%FW)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
LMA (g·m ⁻²)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Lignin (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Cellulose (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Hemicellulose (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Carbon (%DM)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Nitrogen (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Phenolics (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Phosphorus (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Potassium (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540
Sulfur (mg·g ⁻¹)	0.658	0.768	0.710	0.643	0.659	0.606	0.540

Fig. 3 The performance (R^2) of PLSR models for 13 plant traits using three reflectance sources.

Trait value variability

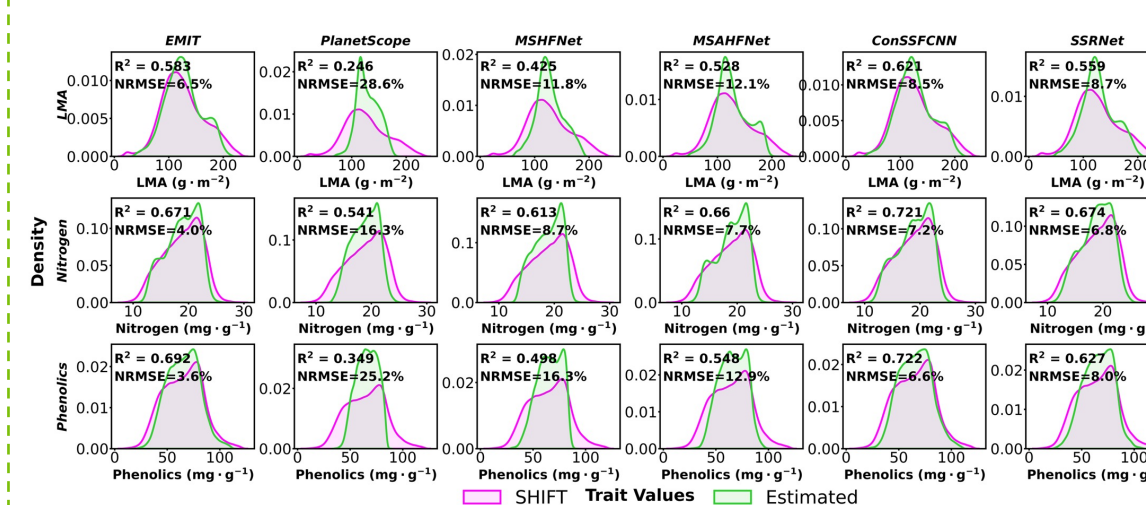


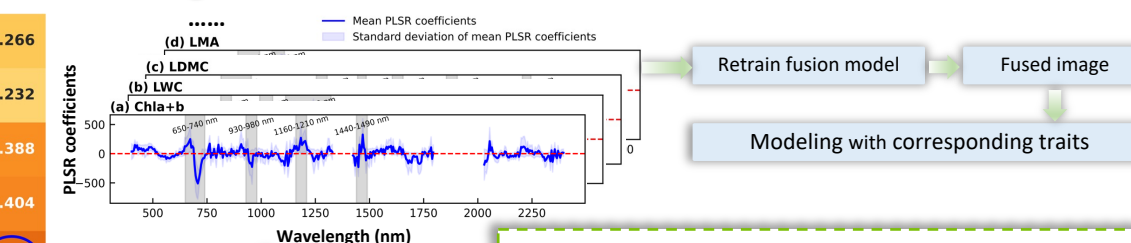
Fig. 5 Comparison of estimated trait value distributions against benchmark trait maps from SHIFT.

Sparse vegetated areas modeling

Reflectance sources	EMIT	PlanetScope	DespNav	MSHNet	MSAHNet	ConSSFCNN	SSRNet
Chla+b (µg·cm ⁻²)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
LWC (mol·m ⁻²)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
LDMC (%FW)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
LMA (g·m ⁻²)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Lignin (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Cellulose (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Hemicellulose (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Carbon (%DM)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Nitrogen (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Phenolics (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Phosphorus (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Potassium (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713
Sulfur (mg·g ⁻¹)	0.216	0.332	0.240	0.507	0.660	0.570	0.713

Fig. 6 The performance (R^2) of PLSR models in sparsely vegetated areas using three reflectance sources.

Features-guided HR-HSI reconstruction for trait estimation



Retraining the fusion model with trait-specific spectral information led to improvements in trait estimation accuracy across most functional traits.

For the traits exhibiting strong sensitivity in the VNIR regions, the fused imagery outperformed EMIT reflectance.

For the traits that rely on absorption features major located in the SWIR, the fused imagery generally underperformed EMIT reflectance.

Take-home messages:

- Fused HR-HSI **substantially improved** trait estimation compared to multispectral imagery, but slightly below EMIT accuracy, while capturing **much finer** spatial heterogeneity.
- In sparsely vegetated areas, **the benefits of fused HR-HSI were trait-dependent**.
- Incorporating trait-specific spectral features into fusion model training **further enhanced accuracy**, particularly for SWIR-sensitive traits.
- Reconstructed HR-HSI quality remains sensitive to sensor noise; **fused trait maps should therefore be interpreted with appropriate caution**.